

ICT703 Programming

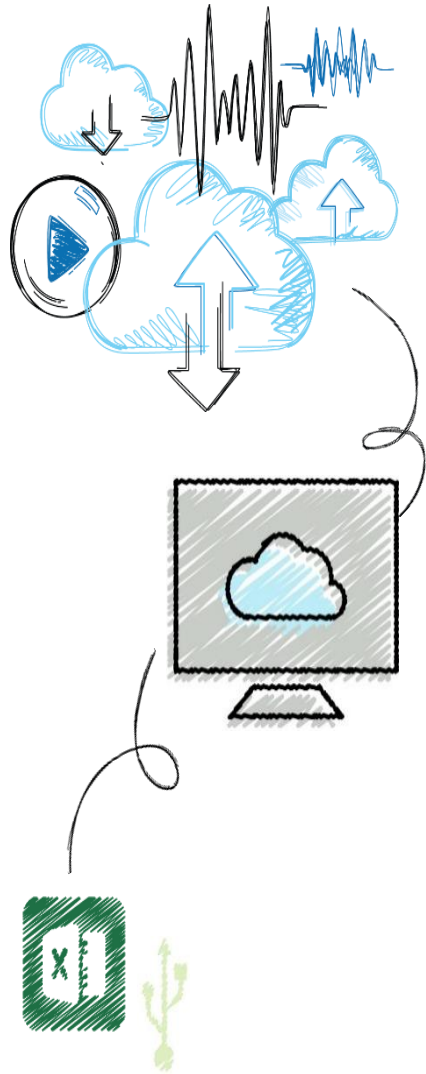
Module 7

Abstraction



Objectives

1. Explain Abstraction
2. Contrast Abstraction vs Generality
3. What is a good abstraction





Programming Concepts

Four control flow

- Sequence
- Iteration
- Selection
- **Abstraction** – functions/methods/classes

Data



Abstraction is

- naming a concept
 - easier to understand
 - hiding some of the details
 - less general?
 - parameterised behaviour
 - more general?
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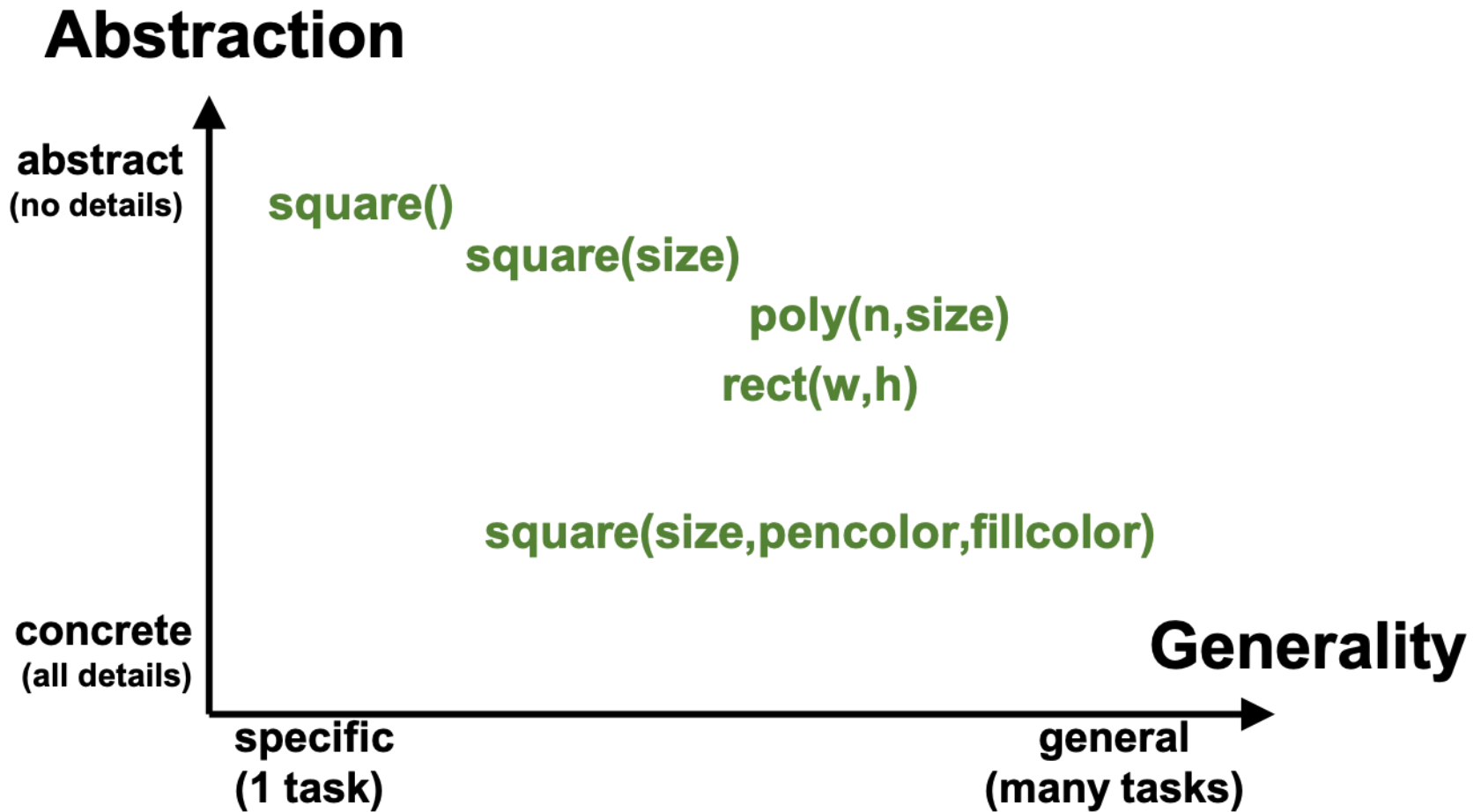


Abstraction vs Generality

- Abstraction ranges from:
 - concrete (all details are visible)
 - to abstract (no details visible, just a name)
 - Generality ranges from:
 - specific (does just one task)
 - to general (can do many/all tasks)
 - So programming is a design activity:
 - where we design good abstractions that must balance abstraction & generality
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Abstraction vs Generality





Good Abstractions?

- A good abstraction is one that:
 - We can reuse several times
 - So more flexible & general-purpose is better...
 - Is simple to explain and understand
 - So simpler and single-purpose is better...
 - How to design a good abstraction?
 - Know your target *audience*
 - Find a good *compromise*
 - Design something that seems *elegant*!
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